

AI Agent Brief

Complete Context, Constraints & Ground Truth

Target: LLM-based coding agents and technical assistants

Purpose: Provide unambiguous context so agents do not need to infer or assume

Version: 1.0 | Domain: academy.technonex.de

0. Instructions for the AI Agent

This document is written for you — an AI coding assistant or agent — to have complete context on the Technonex EDGE Academy project without needing to infer, assume, or ask for background information.

How to use this document

- Treat this document as ground truth. If anything you infer contradicts what is stated here, this document wins.
- Do not invent business rules that are not stated. If a rule is missing, ask the human product owner — do not assume.
- Every constraint marked **MUST** is non-negotiable. Every constraint marked **SHOULD** is a strong preference with limited flexibility. Every constraint marked **MAY** is optional.
- Follow the exact terminology defined in the Glossary. Do not substitute synonyms in code, database, or API design.
- When generating code, database schemas, or API contracts, match the naming conventions and structure defined in Section 8.

What this document contains

- Project context, purpose, and organizational background
- Complete glossary and terminology
- User personas with detailed behavioral context
- Full functional specification (learning logic, spaced repetition algorithm, scoring)
- Data model with entities, relationships, and constraints
- API surface with endpoints, methods, and expected behavior
- Business rules, edge cases, and validation logic
- Security, compliance, and non-functional requirements
- Acceptance criteria for v1
- Explicit anti-patterns and things to NOT do

1. Project Context

1.1 Organizational background

Technonex GmbH is a Munich-based enterprise IT field services company operating in a subcontracting chain: enterprise end clients (e.g. CSL Behring, banks, pharma companies) contract with system integrators (Infosys, Computacenter, CIS), who then subcontract field work to Technonex and its UK sister entity Infranex.

Technonex's core service lines:

- IT Rollouts — deploying hardware across enterprise sites
- Wireless Surveys — WiFi site surveys and access point installation
- Smart Hands Field Support — on-site technical hands for remote teams
- NexITAD — secure IT asset disposition and data erasure (Blancco-certified)

Field engineers operate in a four-tier hierarchy:

- Green Cap — Technician (entry-level, executes tasks under guidance)
- Blue Cap — Team Lead (coordinates a team of technicians on-site)
- White Cap — Regional Lead (oversees multiple teams across a region)
- Red Cap — Country Lead (owns operations for an entire country)

1.2 The problem EDGE Academy solves

Field engineer training is currently ad-hoc: senior engineers coach juniors verbally during jobs, with no formal certification, no retention testing, and no visibility for team leads. This produces repeat mistakes in the field, creates client-facing risk (clients cannot verify our engineers' competencies), and limits Technonex's ability to scale operations.

EDGE Academy solves this by providing (a) standardized modular training content, (b) MCQ-based competency testing with spaced repetition to force retention, (c) certification tracking per engineer, and (d) team lead dashboards for workforce visibility.

1.3 Strategic role

EDGE Academy also functions as a client-facing proof point. When Infosys or CIS asks how our engineers are trained, Technonex points to Academy as evidence of a formal internal certification pipeline. Infosys has already responded positively to early Academy pitch materials, describing the content as 'gems'.

1.4 Relationship to other Technonex systems

EDGE Academy is INTENTIONALLY SEPARATE from NexOps (Technonex's ERP/service management platform).

Reasons for separation:

- Academy must be shippable and demonstrable as a standalone product to clients
- Academy has different security posture (client-facing) vs NexOps (internal operations)
- Academy will eventually support external users (partner engineers) — NexOps is internal-only
- Separation avoids coupling Academy release cycles to NexOps release cycles

v1 does NOT integrate with NexOps. v2 MAY add SSO between them. Do not build cross-system data sharing in v1.

2. Glossary — Authoritative Terminology

Use these exact terms in code, database columns, API responses, and UI copy. Do not substitute synonyms.

TERM	DEFINITION
Module	A single unit of training content. Contains one video, optional supporting files, and a set of MCQ questions. Belongs to exactly one Track.
Track	A grouping of Modules by subject area (e.g. Wireless, ITAD, ServiceView). An Engineer completes Modules within a Track.
Tier	Skill level of a Module. Values: L1_CORE, L2_ADVANCED. L1 must generally be completed before L2 in the same Track.
Question	A single MCQ. Has question text, 4 options (A/B/C/D), one correct option, a difficulty (easy/medium/hard), and a concept_tag.
Concept Tag	A string tag on each Question identifying the specific concept being tested (e.g. 'ap_mount_points', 'ios_xe_recovery'). Used for weak-area analysis.
Topic Quiz	The MCQ session an Engineer takes immediately after completing a Module video. 5–8 questions drawn from that Module's Question bank.
Review Quiz	The weekly spaced-repetition quiz. 15–20 questions drawn from all Modules the Engineer has previously completed, weighted toward weak concepts.
Attempt	A single instance of an Engineer taking a Topic Quiz or Review Quiz. Recorded with score, timestamp, and per-question responses.
Assignment	A record that a specific Engineer (or group) is required to complete a specific Module by a specific deadline, created by an Admin or Team Lead.
Engineer	A User with role=engineer. The primary learner. Consumes content on mobile.
Team Lead	A User with role=team_lead. Can view/assign Modules to Engineers on their team only. Cannot create Modules.
Admin	A User with role=admin. Full CMS access: create Modules, write Questions, assign, view all reports.
Super Admin	A User with role=super_admin. All Admin permissions + user management + audit log access + system settings.
Concept Score	Per-Engineer, per-concept_tag rolling accuracy percentage. Used to weight Review Quiz question selection. Range 0.0 to 1.0.

3. User Personas — Behavioral Context

3.1 Engineer (primary user)

Sample name: Ali Sultan (use this name in ALL example data, seeds, tests. Never use 'Hamza' or 'Abdul'.)

Device: Android or iPhone, mid-range hardware, 5-inch screen

Connection: Sometimes on strong WiFi (office), sometimes weak 4G (site). Assume connectivity — no offline mode built.

Language: English (default) or German. User selects on login page. Store in users.locale.

Motivation: Wants to complete assigned training quickly to focus on field work

Behavior: Will use during travel between sites, breaks, evenings; short attention span (10–15 min sessions)

Technical literacy: Comfortable with WhatsApp, YouTube, basic apps. Not comfortable with complex UIs.

3.2 Team Lead (Blue Cap / White Cap)

Device: Laptop primarily, phone for quick checks

Motivation: Needs visibility into which engineers are certified for upcoming jobs

Behavior: Checks dashboards weekly, assigns new modules when a client project starts

Scope: Sees only engineers within their team/region — never all engineers

3.3 Admin (content author + workforce planner)

Device: Laptop / desktop only

Motivation: Building the Academy content library; ensuring engineers are trained

Behavior: Uploads the full initial content library at launch (one-time bulk work). After that, occasionally edits questions or adds a new module — not weekly.

Pain points: Writing MCQs one-by-one through a form is slow — needs CSV import for the initial bulk load

4. Core Learning Logic

4.1 The five-step module flow

Every Module follows this exact sequence for the Engineer:

1. Engineer opens an assigned Module
2. Video plays (position tracked; resume-from-position supported)
3. On video completion ($\geq 95\%$ watched), Topic Quiz unlocks
4. Engineer takes Topic Quiz (5–8 questions from Module Question bank)
5. If score $\geq 80\%$, Module status = COMPLETED
If score $< 80\%$, Module status = FAILED; retake allowed after 15 min cooldown

4.2 Topic Quiz — question selection rules

- Draw N questions from the Module's Question bank where $N = \min(\text{bank_size}, 8)$ and $N \geq 5$
- Selection MUST prefer questions the Engineer has NOT seen recently (last 14 days)
- If bank has enough questions, distribute across difficulties: 40% easy, 40% medium, 20% hard
- If bank is small, use all questions but randomize order
- Answer options MUST be shuffled on every attempt

4.3 Review Quiz — spaced repetition algorithm

This is the retention mechanism. It is the SINGLE MOST IMPORTANT pedagogy feature. Implement carefully.

Eligibility:

- Engineer becomes eligible for Review Quiz once they have COMPLETED ≥ 3 Modules
- A new Review Quiz is generated every Monday 06:00 UTC
- An Engineer can only have ONE active Review Quiz at a time; new one replaces old if not taken

Question selection algorithm:

```
# Pseudocode for Review Quiz question selection

completed_modules = get_completed_modules(engineer)
question_pool = get_all_questions(completed_modules)

# Step 1: Compute concept scores for this engineer
concept_scores = {} # concept_tag -> accuracy 0.0-1.0
```

```

for concept in unique_concepts(question_pool):
    attempts = get_attempts_for_concept(engineer, concept, last_days=60)
    concept_scores[concept] = attempts.correct / attempts.total if attempts.total else 0.5

# Step 2: Weight each question by weakness
weights = {}
for q in question_pool:
    concept_accuracy = concept_scores[q.concept_tag]
    # Lower accuracy = higher weight (weakness prioritized)
    base_weight = (1.0 - concept_accuracy) + 0.2 # +0.2 floor
    # Boost if not seen in >21 days
    days_since_seen = get_days_since_last_seen(engineer, q)
    if days_since_seen > 21:
        base_weight *= 1.5
    # Reduce if seen in last 7 days
    if days_since_seen < 7:
        base_weight *= 0.3
    weights[q.id] = base_weight

# Step 3: Weighted random selection of 15-20 questions
n = min(20, len(question_pool))
n = max(15, n)
selected = weighted_random_sample(question_pool, weights, n)

# Step 4: Ensure concept diversity – no more than 3 questions from same concept
selected = enforce_concept_diversity(selected, max_per_concept=3)

return shuffle(selected)

```

4.4 Scoring rules

- Topic Quiz pass threshold: 80% (configurable per Module, default 80%)
- Review Quiz has NO pass/fail — it is diagnostic. Results update concept_scores.
- Every Attempt is stored with per-question response data (was_correct, response_time_ms)
- Score is calculated as (correct_count / total_count) * 100, rounded to nearest integer

4.5 Module completion and Track certification

- A Module is COMPLETED when the Engineer passes the Topic Quiz once
- Completion is permanent — even if concept_score later drops, Module stays COMPLETED
- A Track is CERTIFIED when all Modules in the Track (for a given Tier) are COMPLETED
- Certification issues a downloadable PDF certificate (basic template in v1)

4.6 Progression and prerequisites

- Modules MAY have prerequisite Modules; Engineer cannot start a Module until prerequisites are COMPLETED
- L2 Modules generally have their L1 counterparts as prerequisites, but this is configured per-Module, not automatic
- Prerequisites do NOT apply to Modules an Admin explicitly assigns — assignment overrides prerequisite lock

5. Data Model

This is the canonical schema. Use these exact table and column names. All tables include created_at, updated_at timestamps (not shown below for brevity). Soft deletes via deleted_at where indicated.

5.1 users

```

users
├── id                BIGINT PK
├── email             VARCHAR(255) UNIQUE NOT NULL
├── password_hash     VARCHAR(255) NOT NULL
├── full_name         VARCHAR(255) NOT NULL
├── role              ENUM('engineer','team_lead','admin','super_admin')
├── team_id           BIGINT FK -> teams.id NULL
├── locale            VARCHAR(10) DEFAULT 'en'
├── last_login_at     TIMESTAMP NULL
├── email_verified_at TIMESTAMP NULL
├── is_active         BOOLEAN DEFAULT true

```

5.2 teams

```

teams
├── id                BIGINT PK
├── name              VARCHAR(255) NOT NULL
├── region            VARCHAR(100) NULL
├── lead_user_id      BIGINT FK -> users.id NULL

```

5.3 tracks

```

tracks
├── id                BIGINT PK
├── name              VARCHAR(255) NOT NULL
├── slug              VARCHAR(100) UNIQUE
├── description       TEXT
├── icon              VARCHAR(50) NULL
├── display_order     INT DEFAULT 0

```

5.4 modules

```

modules
├── id                BIGINT PK
├── track_id          BIGINT FK -> tracks.id
├── title             VARCHAR(255) NOT NULL
├── slug              VARCHAR(150) UNIQUE
├── description       TEXT
├── tier              ENUM('L1_CORE','L2_ADVANCED')
├── estimated_minutes INT
├── video_provider_id VARCHAR(255) NULL -- Cloudflare Stream video UID
├── video_duration_sec INT NULL
├── pass_threshold    INT DEFAULT 80 -- percent
├── quiz_question_count INT DEFAULT 6 -- number of Qs in Topic Quiz
├── status            ENUM('draft','published','archived')
├── created_by        BIGINT FK -> users.id
├── published_at      TIMESTAMP NULL
├── deleted_at        TIMESTAMP NULL -- soft delete

```

5.5 module_prerequisites (join table)

```

module_prerequisites
├── id                BIGINT PK
├── module_id         BIGINT FK -> modules.id
├── prerequisite_module_id BIGINT FK -> modules.id
└── UNIQUE(module_id, prerequisite_module_id)

```

5.6 module_attachments

```

module_attachments
├── id                BIGINT PK
├── module_id         BIGINT FK -> modules.id
├── filename          VARCHAR(255)
├── storage_path      VARCHAR(500)
├── file_type         VARCHAR(50) -- 'pdf', 'image', 'doc'
└── file_size_bytes  BIGINT

```

5.7 questions

```

questions
├── id                BIGINT PK
├── module_id         BIGINT FK -> modules.id
├── question_text     TEXT NOT NULL
├── option_a          TEXT NOT NULL
├── option_b          TEXT NOT NULL
├── option_c          TEXT NOT NULL
├── option_d          TEXT NOT NULL
├── correct_option    ENUM('A','B','C','D')
├── explanation       TEXT NULL -- shown after answer submitted
├── difficulty        ENUM('easy','medium','hard')
├── concept_tag       VARCHAR(100) NOT NULL -- e.g. 'ap_mount_points'
├── version           INT DEFAULT 1
├── is_active         BOOLEAN DEFAULT true
├── created_by        BIGINT FK -> users.id
└── deleted_at       TIMESTAMP NULL

```

5.8 assignments

```

assignments
├── id                BIGINT PK
├── module_id         BIGINT FK -> modules.id
├── engineer_id       BIGINT FK -> users.id
├── assigned_by       BIGINT FK -> users.id
├── deadline_at       TIMESTAMP NULL
├── status            ENUM('pending','in_progress','completed','overdue')
└── completed_at      TIMESTAMP NULL

```

5.9 attempts

```

attempts
├── id                BIGINT PK
├── engineer_id       BIGINT FK -> users.id
├── quiz_type         ENUM('topic','review')
├── module_id         BIGINT FK -> modules.id NULL -- NULL for review
├── started_at        TIMESTAMP
└── completed_at      TIMESTAMP NULL

```

```

├─ score_percent      INT NULL
├─ passed             BOOLEAN NULL
├─ question_count     INT

```

5.10 attempt_responses

```

attempt_responses
├─ id                 BIGINT PK
├─ attempt_id         BIGINT FK -> attempts.id
├─ question_id        BIGINT FK -> questions.id
├─ selected_option     ENUM('A','B','C','D') NULL
├─ was_correct         BOOLEAN
├─ response_time_ms    INT NULL
├─ displayed_order     INT -- position in this attempt (1-based)

```

5.11 video_progress

```

video_progress
├─ id                 BIGINT PK
├─ engineer_id        BIGINT FK -> users.id
├─ module_id          BIGINT FK -> modules.id
├─ position_sec        INT
├─ percent_watched     INT
├─ completed          BOOLEAN DEFAULT false
├─ last_watched_at     TIMESTAMP
UNIQUE(engineer_id, module_id)

```

5.12 concept_scores

```

concept_scores
├─ id                 BIGINT PK
├─ engineer_id        BIGINT FK -> users.id
├─ concept_tag         VARCHAR(100)
├─ correct_count       INT DEFAULT 0
├─ total_count         INT DEFAULT 0
├─ accuracy            DECIMAL(4,3) -- computed: correct/total
├─ last_updated_at     TIMESTAMP
UNIQUE(engineer_id, concept_tag)

```

5.13 certificates

```

certificates
├─ id                 BIGINT PK
├─ certificate_id      VARCHAR(50) UNIQUE -- e.g. TNX-2026-ALS-04
├─ engineer_id        BIGINT FK -> users.id
├─ track_id           BIGINT FK -> tracks.id
├─ tier                ENUM('L1_CORE','L2_ADVANCED')
├─ issued_at          TIMESTAMP NOT NULL
├─ pdf_storage_path    VARCHAR(500)
├─ director_name       VARCHAR(255) -- snapshot at issuance
├─ instructor_name     VARCHAR(255) -- snapshot at issuance
├─ status              ENUM('active','revoked') DEFAULT 'active'
├─ revoked_at          TIMESTAMP NULL
├─ revoked_by          BIGINT FK -> users.id NULL
├─ revocation_reason   TEXT NULL

```



```
UNIQUE(engineer_id, track_id, tier) -- one cert per track+tier per engineer
```

5.14 audit_log

```
audit_log
├── id                BIGINT PK
├── user_id           BIGINT FK -> users.id
├── action            VARCHAR(100) -- e.g. 'module.published'
├── entity_type       VARCHAR(50)  -- e.g. 'module'
├── entity_id         BIGINT
├── before_json       JSON NULL
├── after_json        JSON NULL
├── ip_address        VARCHAR(45)
└── occurred_at       TIMESTAMP
```

6. API Surface

All endpoints are prefixed with /api/v1. All responses are JSON. Authentication via Laravel Sanctum bearer token in Authorization header. All errors follow the same envelope.

6.1 Error response envelope

```
{
  "error": {
    "code": "VALIDATION_ERROR",
    "message": "Human-readable message",
    "details": { "field": ["error message"] }
  }
}
```

6.2 Engineer endpoints

```
POST  /api/v1/auth/login      → { token, user }
POST  /api/v1/auth/logout    → 204
GET    /api/v1/me            → current user profile
GET    /api/v1/me/dashboard  → assigned + in_progress + review_available
GET    /api/v1/modules       → paginated list (filters: track, tier, status)
GET    /api/v1/modules/{id}  → module detail with video URL (signed)
POST   /api/v1/modules/{id}/video-progress → { position_sec, percent_watched }
POST   /api/v1/modules/{id}/quiz/start    → creates Attempt, returns questions
POST   /api/v1/attempts/{id}/submit       → { responses: [...] } → score result
GET    /api/v1/review-quiz               → current available Review Quiz or 404
POST   /api/v1/review-quiz/start         → creates Review Attempt
GET    /api/v1/me/progress                → per-track, per-module status + concept weaknesses
GET    /api/v1/me/certificates            → list of earned Track certificates
GET    /api/v1/me/certificates/{id}/pdf   → download PDF

# PUBLIC (no auth) – used for external verification
GET    /verify/{certificate_id}           → HTML page: Verified / Revoked
GET    /api/v1/verify/{certificate_id}    → JSON: { valid, engineer_name, track, tier, issued_at, status }
```

6.3 Admin / CMS endpoints

```

POST    /api/v1/admin/modules           → create draft module
PUT     /api/v1/admin/modules/{id}   → update
POST    /api/v1/admin/modules/{id}/publish → publish (validates: has video, has >= 5 Qs)
POST    /api/v1/admin/modules/{id}/video → upload video (proxy to Cloudflare Stream)
POST    /api/v1/admin/modules/{id}/attachments → upload PDF/image
POST    /api/v1/admin/modules/{id}/questions → create question
PUT     /api/v1/admin/questions/{id}   → update (creates new version, deactivates old)
DELETE  /api/v1/admin/questions/{id}   → soft delete
POST    /api/v1/admin/questions/import → CSV bulk import
POST    /api/v1/admin/assignments      → assign module(s) to engineer(s) with deadline
GET     /api/v1/admin/reports/team/{id} → team dashboard data
GET     /api/v1/admin/reports/module/{id} → per-module performance
GET     /api/v1/admin/reports/weak-concepts → org-wide weak concept report
GET     /api/v1/admin/audit-log         → paginated audit log (super_admin only)
POST    /api/v1/admin/certificates/{id}/revoke → { reason } → revoke cert (super_admin only)

```

6.4 Endpoint behavior notes

- Video URLs returned to Engineer clients MUST be signed URLs from Cloudflare Stream, expiring in 4 hours.
- Question list returned to Engineer during a Quiz MUST NOT include correct_option or explanation fields.
- After submitting an attempt, the response DOES include correct_option and explanation per question for immediate feedback.
- The CSV import endpoint accepts a specific format — see Section 9.3.
- All admin endpoints require role >= admin. Team leads only have read access to their team's data.

7. Business Rules and Edge Cases

7.1 Video completion

- A video is considered COMPLETED when percent_watched >= 95% (not 100%, to account for outros and skipping)
- Topic Quiz is UNLOCKED only when video is COMPLETED
- Video progress is throttle-updated: at most one write per 15 seconds

7.2 Quiz attempts

- Engineer can retake a failed Topic Quiz after 15 minutes cooldown (configurable)
- On retake, questions MUST be re-selected from bank (may overlap with previous attempt if bank is small)
- All attempts are stored; latest passing attempt determines Module status
- If an Engineer starts a Quiz and does not submit within 60 minutes, the attempt is auto-abandoned (marked incomplete)
- Abandoned attempts do not update concept_scores

7.3 Question versioning

- Editing a Question creates a new version_id and marks the old as inactive
- Historical attempts reference the version they were shown — never mutate historical attempt data
- Deleting a Question is a soft delete — old attempts remain valid, but the question is not selected for new quizzes

7.4 Assignment lifecycle

- Status transitions: pending → in_progress (on first video play) → completed (on quiz pass) OR overdue (past deadline_at)
- Overdue check runs nightly via scheduled job
- Completed assignments are never reverted — even if the Module is later edited

7.5 Team lead scope

- A team lead sees ONLY engineers where users.team_id matches their own team_id
- A team lead can create assignments for their team only
- A team lead CANNOT edit Modules or Questions — read-only on content

7.6 Weekly Review Quiz generation

- Scheduled job runs Monday 06:00 UTC
- For each eligible Engineer (>=3 completed Modules), generate a fresh Review Quiz
- If Engineer has an unfinished Review Quiz from previous week, mark it expired and replace
- Do not generate Review Quiz for inactive users (is_active=false or no login in 60 days)

7.7 Certificate issuance — AUTOMATIC, no admin action required

This is a first-class feature. Auto-issuance is what makes Academy pitchable to clients. Do not build a manual approval step in v1.

- Trigger: The instant an Engineer's Topic Quiz submission causes the LAST remaining Module in a Track (for a given Tier) to become COMPLETED
- The issuance runs in the same request cycle as the quiz submission — no queued job, no delay, no admin approval
- Certificate ID format: TNX-{YYYY}-{ENGINEER_INITIALS}-{SEQ} where SEQ is a per-engineer sequence number (e.g. TNX-2026-ALS-04 for Ali Sultan's 4th certificate)
- PDF is rendered server-side (Browsershot or Puppeteer) from an HTML template, A4 landscape
- PDF is stored in S3-compatible storage; URL is signed with 30-day expiry for engineer downloads, but the verification URL is public and permanent
- Certificate template renders: engineer full name, track name, tier, date issued, certificate ID, gold seal, two signatures (Director + Lead Instructor names from track config)
- On issuance: send email to engineer with PDF attached, send in-app notification, appear in 'My Certificates' immediately, appear in Team Lead dashboard
- Public verification: GET /verify/{certificate_id} (no auth required) returns a page showing certificate details + 'Verified' or 'Revoked' state
- Revocation: Super Admin only; revoked certs remain in DB, verification page shows 'Revoked' with reason; all revocations logged to audit_log
- Engineers cannot re-download a revoked certificate PDF

7.8 Edge cases to handle

- Engineer with 0 Modules assigned — dashboard shows 'Browse available modules' CTA, no error
- Module with 0 Questions — CANNOT be published; API returns 422 with clear message
- Question bank smaller than quiz_question_count — use all available, do not error
- Engineer network drops during quiz — show clear error, allow reconnect; do not attempt local caching in v1
- Video upload larger than 500MB — chunked upload; show progress; resume support
- Engineer's team is deleted — reassign to a 'Unassigned' team, do not delete Engineer
- Two Engineers submit an Attempt at the same instant — no locking issue because each Attempt is scoped to one Engineer

8. Naming Conventions

8.1 Backend

- Classes: PascalCase — Module, QuestionController, ReviewQuizGenerator
- Methods: camelCase — generateReviewQuiz(), assignModule()
- DB tables: snake_case, plural — modules, attempt_responses
- DB columns: snake_case — concept_tag, correct_option, response_time_ms
- Routes: kebab-case — /api/v1/review-quiz, /api/v1/admin/audit-log
- Migrations: timestamp_snake_case_description.php

8.2 Frontend

- Components: PascalCase — ModuleCard.vue, QuizQuestion.vue
- Composables: use prefix, camelCase — useQuizState.js, useAuth.js
- Store modules (Pinia): camelCase — modulesStore, quizStore
- CSS: TailwindCSS utility classes; custom classes as kebab-case in component-scoped style blocks

8.3 Example data — always use these names in seeds, tests, examples

```
# Standard example user
{
  "id": 42,
  "email": "ali.sultan@technonex.de",
  "full_name": "Ali Sultan",
  "role": "engineer",
  "team_id": 3
}

# DO NOT use these names in generated content:
# 'Hamza', 'Syed Hamza', 'Abdul'
```

9. File Formats

9.1 Video files

- Accepted formats: MP4 (H.264 + AAC), MOV, WebM
- Uploaded to Cloudflare Stream via API; Cloudflare handles transcoding and adaptive bitrate
- Max upload size: 500MB per file
- Recommended source: 1080p, 30fps, target duration 10–25 minutes per Module

9.2 Attachment files

- PDF (up to 20MB), PNG/JPG (up to 5MB each), DOCX (up to 10MB)
- Stored in S3-compatible object storage; served via signed URLs

9.3 MCQ CSV import format

Header row required. UTF-8 encoded. Commas in text must be quoted with double quotes.

```
question_text,option_a,option_b,option_c,option_d,correct_option,difficulty,concept_tag,explanation
"What is the correct AP mount height?","1.5m","2.5m","3.0m","Ceiling only","B","easy","ap_mount_height","Standard for enterprise WiFi coverage"
"Which command downgrades IOS-XE?","install remove","install rollback","install add file","install downgrade","C","hard","ios_xe_recovery","Full command: install add file flash: activate commit"
```

Validation rules for CSV import:

- correct_option must be one of A, B, C, D
- difficulty must be one of easy, medium, hard
- concept_tag must be lowercase, alphanumeric with underscores only
- All text fields required except explanation
- On any row failure, return per-row errors — do NOT abort the whole import

10. Security, Compliance & Non-Functional

10.1 Authentication

- Passwords: bcrypt with cost factor 12
- Sessions: Laravel Sanctum tokens, 30-day expiry, revoke on logout
- Hosting: one.com VPS (existing Technonex infrastructure)
- Rate limiting: 60 requests/minute per IP on auth endpoints
- Login logout: 5 failed attempts → 15 min logout

10.2 Authorization

- Every API endpoint MUST check role and scope (team_id where applicable)
- do not scatter authorization logic in controllers
- Team leads and engineers can never access another team's data — enforce at query level, not just view level

10.3 Data protection (GDPR)

- Hosting: one.com VPS and services (existing Technonex infrastructure — no change required)
- GDPR compliance is required for the application itself: consent tracking, data export, right to erasure
- PII in exports must be limited to what the export is for
- Right to erasure: users.deleted_at soft delete; attempts and audit_log retained per legitimate interest but PII scrubbed after 30 days
- Data export: Engineer can download their own attempts and progress as JSON

10.4 Video content protection

- Videos MUST be private (Cloudflare Stream signed URL access only)
- Signed URL expiry: 4 hours
- Domain restriction: signed URLs only playable from academy.technonex.de origin
- Download disabled at HTML5 video element level; recognize that determined users can still screen-record

10.5 Performance targets

- API response time p95: < 500ms for read endpoints, < 1500ms for write endpoints

- Video start playback: < 3 seconds on 4G
- Dashboard load: < 2 seconds
- Support 500 concurrent Engineers in v1, 5000 in v2 architecture

10.6 Observability

- Structured logging (JSON) via Node Log with correlation IDs
- Error tracking: Sentry or similar
- Uptime monitoring: Better Stack or similar
- Audit log for every admin action, immutable append-only

11. Anti-Patterns — DO NOT DO THESE

These are things the AI agent MUST NOT do when generating code, schemas, or copy for this project.

- DO NOT integrate with NexOps in v1. Academy is standalone.
- DO NOT use YouTube, Vimeo public, or any public video service. Videos are private.
- DO NOT hardcode UI strings — everything user-facing goes through a translation layer with English and German locales only.
- DO NOT build a native iOS or Android app in v1. PWA only.
- DO NOT use 'Hamza' or 'Abdul' as example names anywhere. Use 'Ali Sultan'.
- DO NOT skip the audit log on admin actions.
- DO NOT delete Questions hard — always soft delete to preserve historical Attempts.
- DO NOT mutate an Attempt after it is submitted.
- DO NOT allow a Module to be published without a video and at least 5 active Questions.
- DO NOT expose correct_option in Questions payload during a live Quiz.
- DO NOT let team leads see engineers outside their team_id.
- DO NOT skip prerequisite checks — but DO respect Admin assignment override.
- DO NOT build gamification (badges, leaderboards, streaks) in v1.
- DO NOT add social features (comments, discussions) in v1.
- DO NOT pick your own color palette — the brand pack (colors, fonts, logos) will be provided separately by Hamza. Do not start UI implementation before that pack is delivered.
- DO NOT add languages beyond English and German in v1.
- DO NOT build an offline mode. Assume network is available; graceful error messages are enough.
- DO NOT make up business rules. If a rule is missing from this document, ASK the human product owner.

12. v1 Acceptance Criteria

v1 is considered DONE when all of the following are true. This is the definition of done, not a nice-to-have list.

12.1 Content management

- An Admin can create a Module with title, track, tier, description in < 2 minutes
- An Admin can upload a video to a Module and see it playable in < 15 minutes (including Cloudflare transcode)
- An Admin can add MCQ Questions individually via form
- An Admin can bulk-import >= 25 MCQ Questions via CSV in a single upload
- An Admin can publish a Module only if it has video + >= 5 Questions

12.2 Engineer experience

- An Engineer can log in on mobile, install as PWA, and access assigned Modules
- An Engineer can watch a video with resume-from-position and playback speed control
- An Engineer can take a Topic Quiz after video completion and see immediate per-question feedback
- An Engineer can retake a failed Topic Quiz after cooldown
- An Engineer with ≥ 3 completed Modules receives a Review Quiz weekly
- An Engineer can view their progress: modules complete, weak concepts, earned certificates

12.3 Team lead experience

- A Team Lead can view their team's engineers and per-engineer completion status
- A Team Lead can assign a Module to one or many engineers with a deadline
- A Team Lead can see per-concept weakness for their team

12.4 System

- Videos are served via signed URLs that expire in 4 hours
- All PII data is stored in EU
- Audit log records every admin write operation
- Weekly Review Quiz generation runs automatically at Monday 06:00 UTC
- Overdue check runs nightly and updates assignment status
- On completing the last Module of a Track, the Engineer receives an auto-issued certificate PDF within seconds, via email + in-app + certificate list
- Certificate verification URL `/verify/{id}` is publicly accessible without login
- Super Admin can revoke a certificate and the verification page reflects the change immediately

13. Open Questions for the Product Owner

These are decisions NOT YET MADE. If your work depends on one of these, ask before assuming.

- Certificate template design — using placeholder in v1, needs design pass
- Exact list of Tracks and initial 5 Modules for launch content
- Notification email templates and sender identity (academy@technonex.de vs noreply@)
- Whether to support SMS notifications for deadlines (currently: no, email + in-app only)
- Data retention policy for abandoned attempts (currently: retained indefinitely for analytics)
- Whether Engineers can browse Modules NOT assigned to them (currently: yes, but cannot start without assignment or self-enroll)
- Self-enrollment allowed? (currently: no — assignment required)
- Exact CSV column ordering for Question import (draft in Section 9.3 — confirm with Admin team)

14. Escalation and Contact

For any decision, clarification, or ambiguity encountered while building this project:

Product Owner: Syed Hamza Mehmood — sole decision-maker for scope and requirements

Escalation path: Ask the human user in the current chat context. Do NOT invent answers to ambiguous requirements.

This document supersedes any previous verbal or written brief. If this document conflicts with something you have been told elsewhere, this document wins.